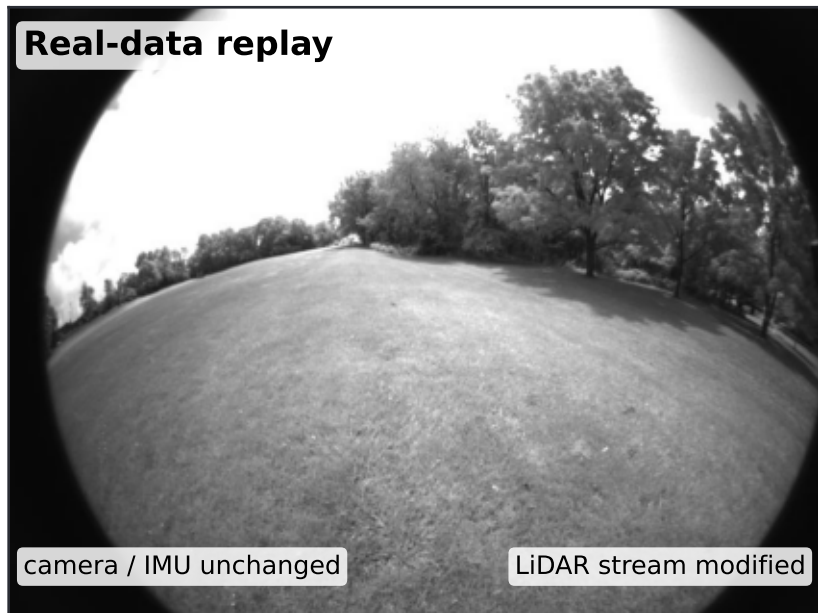
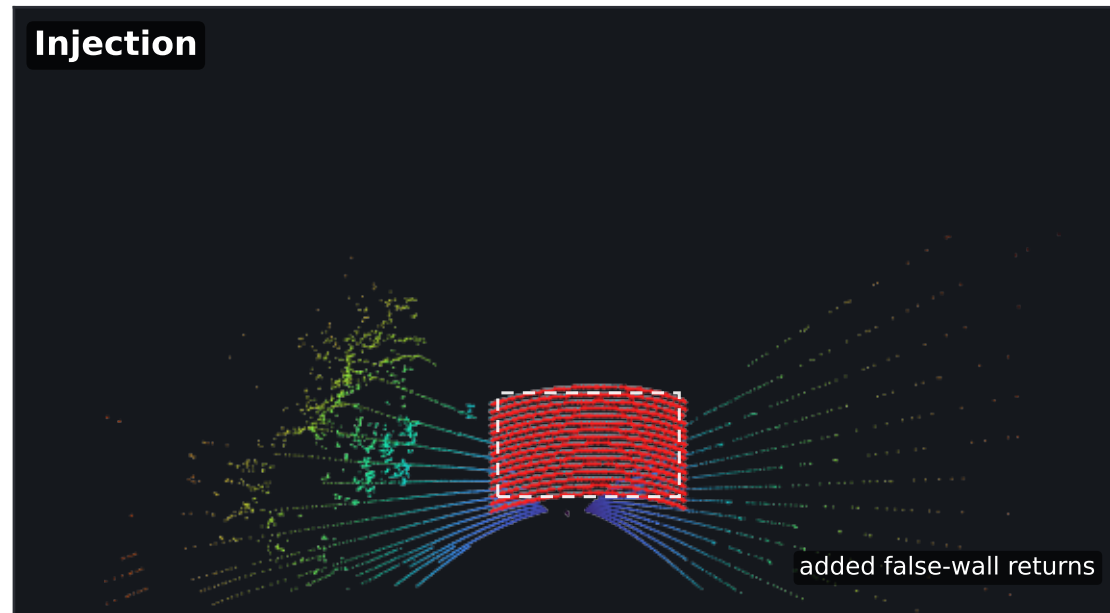


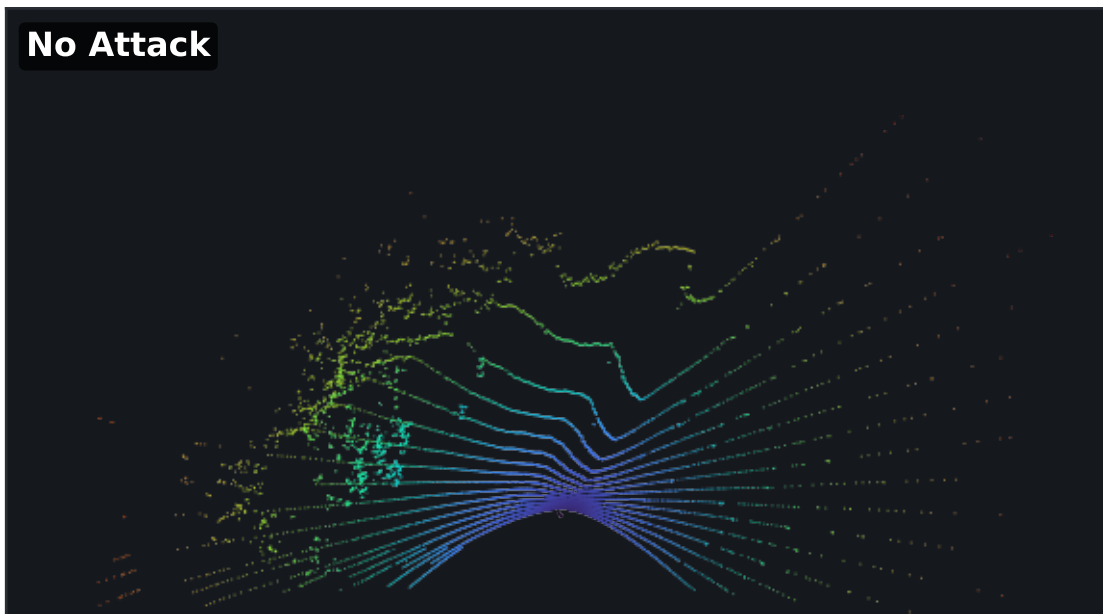
**Real-data replay**



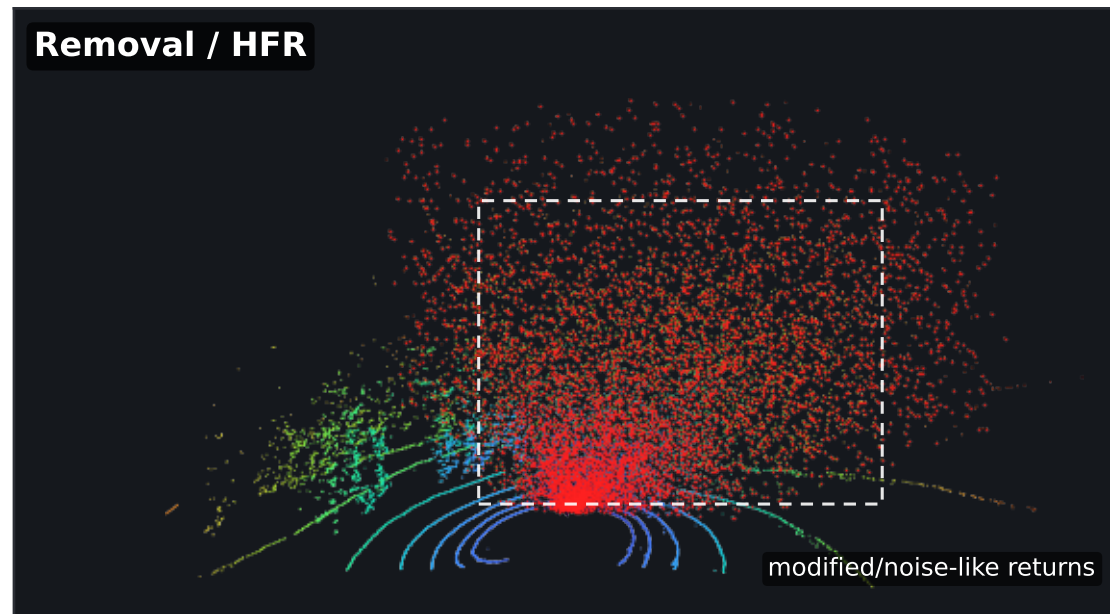
**Injection**



**No Attack**



**Removal / HFR**



Red points denote LiDAR returns introduced or modified by the replayed spoofing model.